

Initialising Quantum States in VQE with a Hartree–Fock Reference

Hybrid quantum–classical methods for eigenvalue problems

Notes for a quantum-computing / quantum-chemistry course

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- 3 Building the reference in code
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The problem we want to solve

For an electronic Hamiltonian $\hat{H}$ acting on N_q qubits we wish to obtain the lowest eigenvalue

$$E_0 = \min_{|\Psi\rangle} \frac{\langle \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle}.$$

Exact diagonalisation scales as $\mathcal{O}(d^3)$ with $d = 2^{N_q}$ — intractable for moderate molecules.

Variational Quantum Eigensolver (VQE) is a *hybrid* algorithm:

- A parameterised circuit $U(\theta) |\Psi_0\rangle$ prepares a trial state on the QPU.
- Expectation values $\langle \Psi(\theta) | \hat{H} | \Psi(\theta) \rangle$ are measured on the QPU.
- A classical optimiser updates θ to minimise the energy.

The variational principle guarantees $E(\theta) \geq E_0$.

Why does the initial state matter?

The ansatz wavefunction is

$$|\Psi(\theta)\rangle = U(\theta) |\Psi_0\rangle.$$

The **reference state** $|\Psi_0\rangle$ controls

- the region of Hilbert space the optimiser can reach with reasonable circuit depth,
- the number of variational parameters needed for chemical accuracy,
- susceptibility to *barren plateaus* and local minima.

A bad reference (e.g. $|0 \cdots 0\rangle$, the qubit vacuum) forces the ansatz to first build in the gross electronic structure before refining correlation.

A good reference encodes the mean-field physics already, so $U(\theta)$ only has to add *electron correlation*.

⇒ The Hartree–Fock state is the natural starting point in quantum chemistry.

Hartree–Fock recap

The Hartree–Fock (HF) method approximates the many-electron wavefunction by a single Slater determinant

$$|\Phi_{\text{HF}}\rangle = \frac{1}{\sqrt{N_e!}} \det[\varphi_{i_1}(x_1) \cdots \varphi_{i_{N_e}}(x_{N_e})],$$

where the spin-orbitals $\{\varphi_p\}$ are obtained by solving the Roothaan–Hall equations self-consistently. In second quantisation, with creation operators $\hat{a}_p^\dagger$ acting on the Fock vacuum $|\text{vac}\rangle$,

$$|\Phi_{\text{HF}}\rangle = \prod_{p \in \text{occ}} \hat{a}_p^\dagger |\text{vac}\rangle$$

The occupied set occ contains the N_e lowest-energy spin-orbitals.

From fermions to qubits

A qubit can represent the occupation of one spin-orbital:

$$|0\rangle_p \leftrightarrow \text{orbital } p \text{ empty}, \quad |1\rangle_p \leftrightarrow \text{orbital } p \text{ occupied}.$$

We need a transformation that maps fermionic operators (with antisymmetry) onto qubit operators (Pauli strings). Standard choices:

- **Jordan–Wigner (JW)** — simple, local in occupation but non-local in parity strings.
- **Parity** mapping — swaps the locality trade-off.
- **Bravyi–Kitaev (BK)** — $\mathcal{O}(\log N)$ locality in both.

Jordan–Wigner:

$$\hat{a}_p^\dagger = \left(\prod_{q < p} Z_q \right) \frac{X_p - iY_p}{2}.$$

Crucially, under JW the HF Slater determinant becomes a single computational-basis state.

The HF state under Jordan–Wigner

With JW ordering (spin-orbitals indexed $0, 1, \dots, M-1$, occupied first),

$$|\Phi_{\text{HF}}\rangle_{\text{JW}} = \underbrace{|1\rangle|1\rangle\cdots|1\rangle}_{N_e \text{ occupied}} \underbrace{|0\rangle|0\rangle\cdots|0\rangle}_{M-N_e \text{ virtual}}.$$

This is a product state! It is prepared from $|0\rangle^{\otimes M}$ by applying X gates on the first N_e qubits:

$$|0\rangle^{\otimes M}: \boxed{X} \quad \boxed{X} \quad \boxed{X} \quad I \quad I \quad \cdots = |\Phi_{\text{HF}}\rangle$$

For H_2 in a minimal basis (STO-3G):

$$M = 4 \text{ spin-orbitals}, \quad N_e = 2 \implies |\Phi_{\text{HF}}\rangle = |1100\rangle.$$

A bare-bones HF initialiser in Qiskit

Goal: prepare $|1100\rangle$ for H_2 with a hand-rolled circuit.

```
from qiskit import QuantumCircuit, QuantumRegister

def hartree_fock_jw(n_spin_orbitals, n_electrons):

    Prepare a Hartree-Fock reference under Jordan-Wigner mapping.
    Occupied orbitals are assumed to be the lowest n_electrons indices.

    qr = QuantumRegister(n_spin_orbitals, name=q)
    qc = QuantumCircuit(qr, name=HF)
    for i in range(n_electrons):
        qc.x(qr[i]) # flip |0> -> |1> on occupied orbitals
    return qc

hf_circuit = hartree_fock_jw(n_spin_orbitals=4, n_electrons=2)
print(hf_circuit)
```

Output (schematic):

q_0: X
q_1: X
q_2:
q_3:

That is the entire HF state preparation under JW: N_o Pauli-X gates.

Using Qiskit Nature's built-in HF initial state

For production work, use the library — it handles JW, parity (with two-qubit reduction), and BK consistently.

```
from qiskit_nature.second_q.circuit.library import HartreeFock
from qiskit_nature.second_q.mappers import JordanWignerMapper

mapper = JordanWignerMapper()

hf_init = HartreeFock(
    num_spatial_orbitals = 2, # H2 in STO-3G has 2 spatial orbitals
    num_particles = (1, 1), # (alpha, beta) electrons
    qubit_mapper = mapper,
)

print(hf_init.draw())
# Equivalent to manually applying X on the occupied spin-orbital qubits.
```

Switching mapper = switching the bit-pattern that represents HF; the library does the bookkeeping for you.

Full pipeline: HF $\rightarrow$ UCCSD $\rightarrow$ VQE for H₂

```
from qiskit_nature.second_q.drivers import PySCFDriver
from qiskit_nature.second_q.mappers import JordanWignerMapper
from qiskit_nature.second_q.circuit.library import HartreeFock, UCCSD
from qiskit_algorithms import VQE
from qiskit_algorithms.optimizers import SLSQP
from qiskit.primitives import Estimator
```

```
# 1. Classical electronic-structure problem (PySCF backend)
driver = PySCFDriver(atom='H 0 0 0; H 0 0 0.735', basis='sto3g')
problem = driver.run()
hamiltonian = problem.hamiltonian.second_q_op()
```

```
# 2. Map fermions -> qubits
mapper = JordanWignerMapper()
qubit_op = mapper.map(hamiltonian)
```

```
# 3. Hartree-Fock initial state
hf_state = HartreeFock(
    num_spatial_orbitals = problem.num_spatial_orbitals,
    num_particles = problem.num_particles,
    qubit_mapper = mapper,
)
```

```
# 4. UCCSD ansatz built ON TOP OF the HF reference
ansatz = UCCSD(
    num_spatial_orbitals = problem.num_spatial_orbitals,
    num_particles = problem.num_particles,
```

What the ansatz actually does

The unitary coupled-cluster singles-and-doubles (UCCSD) ansatz reads

$$|\Psi(\boldsymbol{\theta})\rangle = e^{\hat{T}(\boldsymbol{\theta}) - \hat{T}^\dagger(\boldsymbol{\theta})} |\Phi_{\text{HF}}\rangle, \quad \hat{T} = \hat{T}_1 + \hat{T}_2,$$
$$\hat{T}_1 = \sum_{i \in \text{occ}, a \in \text{virt}} \theta_i^a \hat{a}_a^\dagger \hat{a}_i, \quad \hat{T}_2 = \frac{1}{4} \sum_{ij, ab} \theta_{ij}^{ab} \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i.$$

Two crucial properties when starting from $|\Phi_{\text{HF}}\rangle$:

- At $\boldsymbol{\theta} = 0$, $|\Psi(0)\rangle = |\Phi_{\text{HF}}\rangle$, so the optimiser starts at the HF energy — a known, sensible baseline.
- Each excitation operator $\hat{a}_a^\dagger \hat{a}_i$ promotes an electron from an occupied to a virtual orbital, i.e. it generates correlation *on top of* HF.

Different mappings change the bit pattern

Same physical state, different representation on the qubits:

Mapping	H ₂ HF state	# qubits
Jordan–Wigner	$ 1100\rangle$	4
Parity	$ 0110\rangle$	4
Parity + Z_2 symm.	$ 01\rangle$	2
Bravyi–Kitaev	$ 1110\rangle$	4

Lesson: *never* hard-code an HF bit pattern. Always derive it from the mapper and the orbital ordering, or rely on a library helper.

The two-qubit reduction (parity mapping + Z_2 symmetries) is a particularly useful trick for small molecules — H₂ in STO-3G drops from 4 to 2 qubits.

When HF is not enough

HF is a *mean-field* reference. It breaks down for systems with strong static correlation:

- stretched bonds (H_2 at large R , N_2 dissociation),
- transition-metal complexes with near-degenerate d -orbitals,
- open-shell singlets, conical intersections.

Possible remedies as initial states for VQE:

- **Multi-reference HF** or **CASSCF** $\rightarrow$ linear combination of Slater determinants, prepared with an entangling circuit.
- **Symmetry-broken HF (UHF)** $\rightarrow$ cheaper but loses spin symmetry.
- **Adaptive ansätze** (ADAPT-VQE) $\rightarrow$ grow the ansatz operator-by-operator starting from HF; partly compensates for a poor reference.

A useful diagnostic: if $|1 - \langle S^2 \rangle / S(S+1)|$ is large at $\theta = 0$, HF is a poor starting point.

Preparing a multi-determinant reference

A two-determinant reference, e.g. $|\Psi_0\rangle = \cos \alpha |\Phi_{\text{HF}}\rangle + \sin \alpha |\Phi_{ij}^{ab}\rangle$, needs one entangling block on top of the X-gate layer.

Conceptually:

```
from qiskit import QuantumCircuit
from numpy import pi

def two_det_reference(alpha):
    qc = QuantumCircuit(4)
    # HF part: |1100>
    qc.x(0); qc.x(1)
    # Rotate amplitude into the doubly-excited determinant |0011>
    qc.ry(2*alpha, 0)
    qc.cx(0, 2)
    qc.cx(1, 3)
    qc.x(0); qc.x(1)
    return qc

print(two_det_reference(pi/8).draw())
```

The pattern generalises: any configuration-interaction reference can be prepared with $\mathcal{O}(\# \text{ determinants})$ controlled rotations under JW.

Take-home messages

- VQE turns an eigenvalue problem into an optimisation over circuit parameters — *the starting point of that optimisation matters*.
- The Hartree–Fock state is the natural and cheap reference inherited from classical quantum chemistry.
- Under Jordan–Wigner the HF state is a single product state $|1 \cdots 1 0 \cdots 0\rangle$, prepared by N_e Pauli- X gates.
- Different fermion-to-qubit mappings give different HF bit patterns — let a library compute them for you.
- For strongly correlated systems, supplement HF with a small entangling layer (multi-determinant reference) or use an adaptive ansatz.

Suggested follow-ups:

- Hardware-efficient ansätze and their initialisation pitfalls (barren plateaus).
- Symmetry-preserving ansätze (number, S^2 , S_z conservation).
- Reading: McArdle *et al.*, *Rev. Mod. Phys.* 92, 015003 (2020); Tilly *et al.*, *Phys. Rep.* 986, 1 (2022).

Questions and discussion